

PostgreSQL Subsystem — #d.u.m.b.a.s.s.

Purpose & Scope

PostgreSQL operates as the centralized, persistent relational and vector hub for the entire distributed cognitive ecosystem. It is hosted on **Node Beta** (NVIDIA Jetson Orin Nano Super running Ubuntu Server LTS), bridging edge activity with deep memory and cloud pipelines.

Key Capabilities

1. **migrations_v1.sql**: Initial production migration suite establishing:
 - **repositories**: Registry of federated codebases and active repos.
 - **vector_embeddings**: High-performance semantic memory using **pgvector** with HNSW cosine distance indexing (**vector(1536)**).
 - **audit_ledger**: Complete immutable audit log capturing inputs, outputs, execution latencies, and reward scores (+1.0 / -1.0) from daily driver runs.
 - **quarantine_traces**: Dedicated log for failed scripts and hallucinated trajectories flagged by auditing daemons (including the sealed Red Auditor).
 - **canonical_entities**: Relational knowledge graph linking entities, systems, APKs, and hardware nodes.

Network Topology & Access

- **Host**: Node Beta (192.168.x.x / Tailscale IP) on port **5432**.
- **Clients**:
 - Edge sync daemons on Node Alpha push batched episodic logs at the end of sessions.
 - OmniRoute pulls entity schemas and cached vector contexts during query synthesis.
 - Python scripts in **skills-and-capabilities/** execute semantic clustering and graph validation queries directly.